Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/52**

Paper 5 Probability & Statistics 1 **October/November 2021**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

© UCLES 2021

[Turn over

PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2021

Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

© UCLES 2021

PUBLISHED

Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

© UCLES 2021

Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

© UCLES 2021

Question	Answer	Marks	Guidance
1(a)	$\frac{82}{180}, \frac{41}{90}$, 0.456	B1	
		1	
1(b)	$\left[\mathrm{P(M\|D)} = \frac{P(M \cap D)}{P(D)}\right] = \frac{\frac{11}{180}}{\frac{20}{180}+\frac{11}{180}}$ or $\frac{0.6011}{0.1722}$	M1	*Their* identified $\frac{P(M \cap D)}{P(D)}$ *or* from data table $\frac{11}{20+11}$, accept unsimplified, condone × 180.
	$\frac{11}{31}$, 0.355	A1	Final answer.
		2	

Question	Answer	Marks	Guidance
1(c)	$P(F) = \frac{100}{180}, \frac{5}{9}, 0.5556$ OE $P(G) = \frac{82}{180}, \frac{41}{90} 0.4556$ OE $P(F \cap G) = \frac{38}{180}, \frac{19}{90}, 0.2111$ OE $P(F) \times P(G) = \frac{100}{180} \times \frac{82}{180} = \frac{41}{162}, 0.2531$ OE $\left[\neq \frac{38}{180}\right]$ Not independent	**M1**	*Their* identified P(F) × *their* identified P(G) or correct seen, can be unsimplified.
		A1	$\frac{41}{162}, \frac{38}{180}, P(F \cap G)$ and $P(F) \times P(G)$ seen with correct conclusion, WWW. Values and labels must be seen.
	Alternative method for question 1(c)		
	$P(F \cap G) = \frac{38}{180}, \frac{19}{90}, 0.2111$ OE $P(G) = \frac{82}{180}, \frac{41}{90}, 0.4556$ OE $P(F\|G) = \frac{\frac{38}{180}}{\frac{82}{180}} = \frac{19}{41}, 0.4634$ OE $\neq P(F) = \frac{100}{180}, \frac{5}{9}, 0.5556$ OE Not independent	**M1**	P(F|G) (OE) unsimplified with *their* identified probs or correct
		A1	$\frac{19}{41}, \frac{100}{180}, P(F \cap G)$ and $P(F\|G)$ seen with correct conclusion WWW. Values and labels must be seen.
		2	

Question	Answer	Marks	Guidance
2(a)	$^{11}C_5 \times {}^4C_1$	**M1**	$^{11}C_5 \times {}^4C_1$ condone $^{11}P_5 \times {}^4P_1$ no +, –, × or ÷.
	1848	**A1**	CAO as exact.
		2	
2(b)	**Method 1 [**Identifying scenarios]		
	[Neither selected =] $^{13}C_6$ [= 1716] [Only Jane selected =] $^{13}C_5$ [= 1287] [Only Kate selected =] $^{13}C_5$ [= 1287]	**M1**	Either $^{13}C_6$ seen alone or $^{13}C_5$ seen alone or × 2 (condone $^{13}P_n$, n = 5,6).
	[Total =] 1716 + 1287 + 1287	**M1**	Three correct scenarios only added, accept unsimplified (values may be incorrect).
	4290	**A1**	
	Method 2 [Total number of selections – selections with Jane and Kate both picked]		
	$^{15}C_6$ - $^{13}C_4$ [= 5005 – 715]	**M1**	$^{15}C_6 – k$, k a positive integer < 5005, condone $^{15}P_6$.
		M1	$m – {}^{13}C_4$, m integer > 715, condone $n – {}^{13}P_4$, n > 17 160.
	4290	**A1**	
		3	
			SC Where the condition of **2(a)** is also applied in **2(b)**, the final answer is 1512 **SC M1 M1 A0** max. The method marks can be earned for the equivalent stages in each method. Method 1 $^4C_1 \times {}^9C_5 + {}^4C_1 \times {}^9C_4 \times 2$ Method 2 $^4C_1 \times {}^{11}C_5 – {}^4C_1 \times {}^9C_3$

© UCLES 2021

Question	Answer	Marks	Guidance
3(a)	For one yellow: YGG + GYG +GGY $\frac{5}{9}\times\frac{4}{8}\times\frac{3}{7}\times 3$	**M1**	$\frac{a}{9}\times\frac{b}{8}\times\frac{c}{7}$, $0 < a,b,c$ integers $\leqslant 5$, for one arrangement.
		M1	*Their* three-factor probability $\times$ 3, ${}^3C_1$, ${}^3C_2$ or ${}^3P_1$, (or repeated adding) no additional terms.
	$\left[\frac{180}{504}=\right]\frac{5}{14}$	**A1**	AG. Convincingly shown, including identifying possible scenarios, may be on tree diagram WWW.
		3	
	Alternative method for question 3(a)		
	$\frac{{}^5C_1\times{}^4C_2}{{}^9C_3}$	**M1**	$\frac{{}^5C_1\times{}^4C_2}{{}^9C_r}$, $r = 2, 3, 4$
		M1	$\frac{{}^5C_s\times{}^4C_t}{{}^9C_3}$, $s + t = 3$
	$\left[\frac{30}{84}=\right]\frac{5}{14}$	**A1**	AG. Convincingly shown, WWW.
		3	

Question	Answer	Marks	Guidance
3(b)	X: 0 | 1 | 2 | 3 $P(X)$: $\frac{24}{504}$ $\left[=\frac{1}{21}, 0.0476\right]$ | $\frac{180}{504}$ $\left[=\frac{5}{14}, 0.357\right]$ | $\frac{240}{504}$ $\left[=\frac{10}{21}, 0.476\right]$ | $\frac{60}{504}$ $\left[=\frac{5}{42}, 0.119\right]$	**B1**	Table with correct X values and one correct probability inserted appropriately. Condone any additional X values if probability stated as 0.
		B1	Second identified correct probability, may not be in table.
		B1	All probabilities identified and correct . **SC** if less than 2 correct probabilities or X value(s) omitted: **SC B1** 3 or 4 probabilities summing to one.
		3	
3(c)	[E(X) =] $\frac{840}{504}, \frac{5}{3}$, 1.67	**B1**	OE Must be evaluated. **SC B1 FT** correct unsimplified expression from incorrect **3(b)** using at least 3 probabilities, $0 < p < 1$.
		1	

Question	Answer	Marks	Guidance
4(a)	$\frac{9!}{3!}$	**M1**	$\frac{9!}{e!}$, $e = 2, 3$
	60 480	**A1**	
		2	

Question	Answer	Marks	Guidance
4(b)	$\frac{7!}{3!}\times 2\times 6$	**M1**	$\frac{7!}{3!}\times k$ seen, k an integer > 0.
		M1	$\frac{m!}{n!}\times 2\times q$ $7 \leqslant m \leqslant 9$, $1 \leqslant n \leqslant 3$, $1 \leqslant q \leqslant 8$ all integers.
		M1	$\frac{m!}{n!}\times p\times 6$ $7 \leqslant m \leqslant 9$, $1 \leqslant n \leqslant 3$, $1 \leqslant p \leqslant 2$ all integers. (Accept 3P2 for 6) If **M0 M0 M0** awarded, **SC M1** for $t \times 12$, t an integer $\geqslant 20$, $\frac{5!}{3!}$.
	10 080	**A1**	Exact value.
	Alternative method for question 4(b)		
	$\frac{{}^7\mathrm{P}_2\times 6!\times 2}{3!}$	**M1**	$\frac{6!}{3!}\times k$ seen, k an integer > 0.
		M1	$\frac{m!}{n!}\times {}^7\mathrm{P}_2\times q$ $m = 6,9$, $1 \leqslant n \leqslant 3$, $1 \leqslant q \leqslant 2$ all integers.
		M1	$\frac{m!}{n!}\times {}^7\mathrm{P}_r\times 2$ $m = 6, 9$, $1 \leqslant n \leqslant 3$, $1 \leqslant r \leqslant 5$ all integers. If **M0 M0 M0** awarded, **SC M1** for $t \times 84$, t an integer $\geqslant 20$, $\frac{5!}{3!}$.
	10 080	**A1**	Exact value.

Question	Answer	Marks	Guidance
4(b)	**Alternative method for question 4(b)**		
	$\frac{7!}{3!}\times 4P2$	**M1**	$\frac{7!}{3!}\times k$ seen, k an integer > 0.
		M1	$t\times {}^4P_2$ or 12, t an integer $\geqslant 20$, $\frac{5!}{3!}$.
		M1	$\frac{m!}{n!}\times 4P2$ $7 \leqslant m \leqslant 9$, $1 \leqslant n \leqslant 3$ all integers.
	10 008	**A1**	Exact value.
		4	

Question	Answer	Marks	Guidance
5(a)	[P(0, 1, 2) =] ${}^{10}C_0\, 0.16^0\, 0.84^{10} + {}^{10}C_1\, 0.16^1\, 0.84^9 + {}^{10}C_2\, 0.16^2\, 0.84^8$ [= 0.17490 + 0.333145 + 0.28555]	**M1**	One term: ${}^{10}C_x p^x\, (1-p)^{10-x}$ for $0 < x < 10$, any p.
		A1	Correct unsimplified expression, or better.
	0.794	**A1**	$0.7935 < p \leqslant 0.794$, mark at most accurate. If **M0** scored, **SC B1** for final answer 0.794.
		3	
5(b)	$(0.84)^7\, 0.16$	**M1**	$(1-p)^7 p$, $0 < p < 1$
	0.0472	**A1**	0.0472144 to at least 3sf.
		2	

Question	Answer	Marks	Guidance
5(c)	$4 \times 0.0472 \times (1 - 0.0472)^3$	**M1**	$4 \times q(1-q)^3$, q = *their* **(b)** or correct.
	0.163	**A1**	$0.163 \leqslant p \leqslant 0.1634$, mark at most accurate from *their* probability to at least 3sf.
		2	

Question	Answer	Marks	Guidance
6(a)	$\left[\mathrm{P}(X > 28.6) = \right] P\left(Z > \dfrac{28.6 - 32.2}{9.6}\right)$ $\left[= P(Z > -0.375)\right]$	**M1**	28.6, 32.2 and 9.6 substituted appropriately in ± Standardisation formula once, allow continuity correction of ± 0.05, no σ^2, $\sqrt{\sigma}$.
	$[\Phi(\textit{their}\, 0.375) =]$ *their* 0.6462	**M1**	Appropriate numerical area, from final process, must be probability, expect > 0.5.
	0.646	**A1**	AWRT
		3	
6(b)	$z = \pm 0.842$	**B1**	$0.841 < z \leqslant 0.842$ or $-0.842 \leqslant z < -0.841$ seen.
	$\dfrac{t - 32.2}{9.6} = 0.842$	**M1**	Substituting 32.2 and 9.6 into ± standardisation formula, no continuity correction, allow σ^2, $\sqrt{\sigma}$, must be equated to a z-value.
	$t = 40.3$	**A1**	$40.28 \leqslant t \leqslant 40.3$ WWW
		3	

Question	Answer	Marks	Guidance
6(c)	$P\left(-\frac{15}{9.6} < Z < \frac{15}{9.6}\right)$ $P(-1.5625 < Z < 1.5625)$	**M1**	Identifying at least one of $\frac{15}{9.6}$ and $-\frac{15}{9.6}$ as the appropriate z-values or substituting *their* (32.2 ± 15) into ± Standardisation formula once, no continuity correction, σ^2 nor $\sqrt{\sigma}$. Condone ±1.563 for **M1**.
	$[2\ \Phi(\frac{15}{9.6}) - 1\ ]$ $= 2 \times 0.9409 - 1$	**A1**	$p = 0.941$ AWRT SOI
		M1	Appropriate area $2\Phi - 1$ oe, (eg $1 - 2 \times 0.0591$, $2 \times (0.9409 - 0.5)$ or $0.9409 - 0.0591$), from final process, must be probability > 0.5.
	0.882	**A1**	
		4	

Question	Answer	Marks	Guidance
7(a)	Cumulative frequency graph drawn	**B1**	Axes labelled 'cumulative frequency' (or cf) from 0 to at least 140 and 'distance (or d) [in] m' from 0 to at least 1600, linear scales with at least 3 values stated.
		B1	All plotted correctly at correct upper end points (200 etc.) **curve** drawn accurately joined to (0, 0) (straight line segments B0) but no daylight above 140. Cf scale no less than 2 cm = 20 children .
		2	

Question	Answer	Marks	Guidance
7(b)	[UQ at 75% of 140 = 105, LQ at 25% of 140 = 35] [IQR:] 700 – 260	**M1**	Accept 660 ⩽ UQ ⩽ 720 – 240 ⩽ LQ ⩽ 290. If values are outside our range, FT providing scales linear and increasing cf drawn.
	440	**A1**	Accept correct evaluation of 660 ⩽ *their* UQ ⩽ 720 – 240 ⩽ *their* LQ ⩽ 290 with clear indication that graph has been used for at least one of 105 or 35.
		2	

© UCLES 2021

Question	Answer	Marks	Guidance
7(c)	[Mean =] $\frac{16\times100+30\times250+42\times400+34\times700+12\times1050+6\times1400}{140}$	**B1**	<table><tr><td>Frequencies</td><td>16</td><td>30</td><td>42</td><td>34</td><td>12</td><td>6</td></tr><tr><td>Mid-points</td><td>100</td><td>250</td><td>400</td><td>700</td><td>1050</td><td>1400</td></tr></table> 5 or 6 correct frequency values seen.
		B1	5 or 6 correct midpoint values seen.
		M1	Values substituted into mean formula using *their* midpoints which must be in the class – condone 1 data error. Accept $\frac{1600+7500+16\,800+23\,800+12\,600+8400}{140}$ or $\frac{70\,700}{140}$. Condone $\frac{70\,770}{140}$ for **M1**.
	505	**A1**	WWW
	Variance = $\frac{16\times100^2+30\times250^2+42\times400^2+34\times700^2+12\times1050^2+6\times1400^2}{140}$ -505^2	**M1**	Values substituted into variance formula using (*their* mean)^2 and *their* midpoints and *their* frequencies (including for denominator). Accept unsimplified. Condone 1 data error. Accept: $[\frac{160\,000+1\,875\,000+6\,720\,000+16\,660\,000+13\,230\,000+11\,760\,000}{140}$ or $\frac{50\,405\,000}{140}$ or 360 035.7143] $-\left[505^2 \text{ or } 255\,025\right]$ If formula stated accept 105 010 or 105 011 WWW.
	S.d. = $\left[\sqrt{105\,010.7}=\right]324$	**A1**	WWW
		6	